PostgresSplitBrain

Overview

- This alert `PostgresSplitBrain`, is designed to detect a split-brain scenario in a PostgreSQL cluster managed by Patroni. It validates if each Patroni cluster has only one Primary node ie: each `type` of Patroni cluster only has one Primary node accepting R/W requests.
- An incorrect consul configuration or Patroni configuration can trigger this , Incorrect health check results can cause Patroni to incorrectly promote a new leader or fail to demote a current leader.

Services

- Postgres Overview
- Patroni Service
- Consul
- Team that owns the service: Production Engineering : Database Reliability

Metrics

- Link to the metrics catalogue
- This Prometheus expression counts the number of PostgreSQL instances in the gprd/gstg environment that are not in replica mode `(pg_replication_is_replica == 0)`. If this count is greater than 1, it triggers the alert. This condition indicates that more than one PostgreSQL instance is operating in read-write mode within a cluster.This condition must be true for at least 1 minute to trigger the alert.
- In both `gstg` and `gprd` environments we have 3 patroni clusters (main, ci and registry at the time of writing) and as such we should expect 3 total primary nodes. Be sure to check the `type` field to distinguish the Patroni clusters and ensure there is only 1 per type. If we see more than three different `type` of clusters it might suggest a `PatroniConsulMultipleMaster` situation.

Alert Behavior

- We can silence this alert by going here, finding the `PostgresSplitBrain` and click on silence option
- This is a very rare and critical event
- There might be a sudden spike in the gitlab_schema_prevent_write errors , Link to dashboard

Severities

- This alert might create S1 incidents.

- Who is likely to be impacted by this cause of this alert?
  - Depending on the database it could be some or all customers. If it is the `main` or `ci` database then we expect nearly all customers would be impacted. If it is the `registry` database then it would be a subset of customers that are depending on the registry.
- Review Incident Severity Handbook page to identify the required Severity Level

Verification

- To validate if a Patroni cluster has more than one leader , a quick view through this dashboard should give you that information regarding the `type` of the Patroni cluster which has more than one leader if any

- On executing the query `pg_replication_is_replica{type="<type of cluster with more than one replica>", env="<gstg/gprd>"}` on your grafana dashboard should tell you the cluster and the fqdn of the leader nodes , please note the `pg_replication_is_replica` value for leader nodes would be 0

Recent changes

- Recent Patroni Service change issues

- Recently closed issues to determine, if a CR was completed recently, which might be correlated: Recently Closed Issues

Troubleshooting

- First step is to figure out which Patroni cluster has more than one leader , a quick view through this dashboard should give you that information regarding the `type` of the Patroni cluster which has more than one leader

- On executing the query `pg_replication_is_replica{type="<type of cluster with more than one replica>", env="gprd"}` on your grafana dashboard should tell you the cluster and the fqdn of the leader nodes , please note the `pg_replication_is_replica` value for leader nodes would be 0

- It might be helpful to look in recent MRs to see if any changes rolled out recently related to Patroni clusters in chef repo, or config-mgmt

- Steps that can be used via the Patroni CLI to remediate . For example:

```
#(Untested steps proceed with extreme caution)
# We ssh into one of the replicas
ssh patroni-main-v14-03-db-gprd.c.gitlab-production.internal
# Verify the members in the cluster and find the erroneus leader
sudo gitlab-patronictl list
```

```
# Pausing the incorrect/erroneus leader , this will make Postgres stop on the node
sudo mv /var/opt/gitlab/postgresql/data12 /var/opt/gitlab/postgresql/data"<find the number g
# Once unreachable manually promote a new node to leader
sudo gitlab-patronictl failover
```

- Helpful gitlab-patronictl commands

Possible Resolutions

- Issue 15773

Dependencies

- Internal dependencies like issues with Patroni configuration , Consul configuration , false positive healthchecks of nodes may cause a splitbrain situation
- External dependencies like networ outage or high network latency may also cause a splitbrain situation.

Escalation

- Please use /devoncall on Slack for any escalation that meets the criteria.
- Slack channels where help is likely to be found: `#g_infra_database_reliability`, `#database`

Definitions

- Link to the definition of this alert for review and tuning
- Link to edit this playbook
- Update the template used to format this playbook

Related Links

- Related alerts
- Postgres Runbook docs
- Update the template used to format this playbook